ANSWER KEY & MARKING SCHEME SET 1

Chapter 3: **Metals and Non-metals** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** $Fe + CuSO_4$ solution	Fe is more reactive than Cu; others involve a less reactive metal.
2	**(b)** its easy loss of an electron	Sodium readily forms Na^+.
3	**(c)** electrolysis of molten chlorides	They are too reactive for carbon reduction.
4	**(c)** applying grease	Paint/zinc cannot be used on a pan used for cooking; grease is suitable.
5	**(d)** calcium	Ca reacts less violently and floats.
6	**(b)** impure copper	Impure copper dissolves.
7	**(b)** calcium	A metal forms an ionic oxide; calcium oxide is ionic and dissolves.
8	**(b)** copper	Copper is below hydrogen in the series.
9	**(d)** acidified copper sulphate solution	Cu^{2+} ions move to the cathode.
10	**(c)** zinc is more reactive than tin	Zinc would react with acids in food.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Mercury (1); Sodium (or potassium) (1).	1+1
12	Mixture of conc. HCl and conc. HNO_3 in 3 : 1 (1); It dissolves gold and platinum (1).	1+1
13	$Na \longrightarrow Na^+ + e^-$ and $Cl + e^- \longrightarrow Cl^-$ (1); $Na^+ + Cl^- \longrightarrow NaCl$ (1).	1+1
14	It reacts very vigorously with oxygen and moisture (1); Kerosene keeps it away from air/water preventing fire (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Hammer: metals are malleable, non-metals break into pieces (1); Circuit: metals complete the circuit and the bulb glows (1); Non-metals (except graphite) do not conduct (1).	1+1+1
16	Calcination: $ZnCO_3 \longrightarrow ZnO + CO_2$ (1); Reduction: $ZnO + C \longrightarrow Zn + CO$ (1); Carbon acts as reducing agent (1).	1+1+1
17	Metals conduct heat and electricity; non-metals do not (graphite exception) (1); Metal oxides are basic (1); Non-metal oxides are acidic (1).	1+1+1
	OR: $3Fe + 4H_2O \longrightarrow Fe_3O_4 + 4H_2$ (1); $Ca + 2H_2O \longrightarrow Ca(OH)_2 + H_2$ (1); Calcium floats due to hydrogen bubbles (1).	1+1+1
18	Mercury is a very unreactive metal (1); $2HgS + 3O_2 \longrightarrow 2HgO + 2SO_2$ (1); $2HgO \longrightarrow 2Hg + O_2$ (1).	1+1+1
	OR: Making the oxide layer on aluminium thicker by electrolysis (1); Aluminium is made the anode in dilute sulphuric acid (1); Thicker oxide layer resists corrosion and can be dyed (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$4Na + O_2 \longrightarrow 2Na_2O$ (1); $2Mg + O_2 \longrightarrow 2MgO$ (1); $4Al + 3O_2 \longrightarrow 2Al_2O_3$ (amphoteric) (1); $2Cu + O_2 \longrightarrow 2CuO$ (black) (1); Amphoteric oxides react with both acid and base (1).	1+1+1+1+1
	OR: Slow attack of air and moisture on metals (1); Rusting needs both air and water (1); Painting and oiling (1); Galvanisation and chrome plating (1); Alloying and anodising (1).	1+1+1+1+1
20	Order: Na > Mg > Al > Zn > Fe > Pb > Cu > Ag > Au (1); Na/K react violently with cold water (1); Mg reacts with hot water, Al/Zn/Fe with steam (1); Cu, Ag, Au do not react (1); Metals above hydrogen displace it from acids (1).	1+1+1+1+1
	OR: Lustre, malleability, ductility (1); Good conductors of heat and electricity (1); Sonorous and generally hard (1); Exception: mercury is liquid (1); Exception: sodium, potassium are soft/low melting point (1).	1+1+1+1+1

Mistakes to avoid:

- Storing sodium or potassium in water; they are stored under kerosene because they react violently with water and air.
- Confusing **roasting** (sulphide ore, excess air) with **calcination** (carbonate ore, limited air).
- Writing the product of iron with steam as Fe_2O_3; it is Fe_3O_4 and hydrogen gas is released.
- Thinking that all metals are hard solids and all non-metals are non-conductors; Na is soft, Hg is a liquid and graphite conducts.

- Saying that an ionic solid conducts electricity; it conducts only when molten or in aqueous solution.
- Writing that copper can displace hydrogen from dilute acids or iron from $\mathrm{FeSO_4}$; a less reactive metal cannot displace a more reactive one.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 3: **Metals and Non-metals** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** $MgCl_2$	Metal + non-metal compound is ionic.
2	**(b)** copper and zinc	Brass: Cu + Zn.
3	**(d)** amphoteric oxides	Al_2O_3 and ZnO are amphoteric.
4	**(d)** in molten state and in aqueous solution	Ions are free to move only when molten or dissolved.
5	**(a)** copper and tin	Bronze: Cu + Sn.
6	**(c)** sodium	Sodium reacts vigorously with oxygen and moisture.
7	**(d)** cinnabar	Cinnabar is HgS.
8	**(b)** lead and tin	It has a low melting point.
9	**(b)** mercury	Mercury is the liquid metal.
10	**(a)** bauxite	Bauxite is $Al_2O_3 \cdot 2\,H_2O$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Zinc is more reactive than tin (1); It would react with food acids and spoil the food (1).	1+1
12	Coating iron with a layer of zinc (1); Prevents rusting; zinc protects even when scratched (1).	1+1
13	$Al_2O_3 + 6\,HCl \longrightarrow 2\,AlCl_3 + 3\,H_2O$ (1); $Al_2O_3 + 2\,NaOH \longrightarrow 2\,NaAlO_2 + H_2O$ (1).	1+1
14	Ore: mineral from which metal can be profitably extracted (1); Gangue: impurities like soil and sand in the ore (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Blue colour fades (1); Reddish-brown copper is deposited (1); $Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu$ (1).	1+1+1
16	Displacement of iron from Fe_2O_3 by aluminium (1); $Fe_2O_3 + 2\,Al \longrightarrow Al_2O_3 + 2\,Fe + heat$ (1); Molten iron welds railway tracks/cracked machine parts (1).	1+1+1
17	Good conductors of electricity (1); Copper: cooking vessels (1); Aluminium: foil for packaging (1).	1+1+1
	OR: Mg (2,8,2) loses 2 electrons giving Mg^{2+} (1); Each Cl (2,8,7) gains 1 electron giving Cl^- (1); One Mg combines with 2 Cl: $MgCl_2$ (1).	1+1+1
18	Electrolysis of molten NaCl (1); Cathode: $Na^+ + e^- \longrightarrow Na$ (1); Anode: $2\,Cl^- \longrightarrow Cl_2 + 2\,e^-$ (1).	1+1+1
	OR: $2\,Al + 6\,HCl \longrightarrow 2\,AlCl_3 + 3\,H_2$ (1); Hydrogen (1); It burns with a pop sound (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Reactivity series: metals arranged in decreasing reactivity (1); Iron nail in $CuSO_4$: brown deposit (1); Solution turns green (1); $Fe + CuSO_4 \longrightarrow FeSO_4 + Cu$ (1); Fe is more reactive than Cu (1).	1+1+1+1+1
	OR: Non-lustrous, brittle, non-conductors (1); Low melting/boiling points (1); Exception: diamond is hard and lustrous (1); Exception: graphite conducts electricity (1); Iodine is lustrous (1).	1+1+1+1+1
20	$Na \longrightarrow Na^+ + e^-$, $Cl + e^- \longrightarrow Cl^-$ (1); $Mg \longrightarrow Mg^{2+} + 2\,e^-$ (1); Solid, brittle, hard (1); High melting/boiling points (1); Soluble in water; conduct when molten/dissolved (1).	1+1+1+1+1
	OR: Oxides reacting with both acids and bases (1); $Al_2O_3 + 6\,HCl \longrightarrow 2\,AlCl_3 + 3\,H_2O$ (1); $Al_2O_3 + 2\,NaOH \longrightarrow 2\,NaAlO_2 + H_2O$ (1); $ZnO + 2\,HCl \longrightarrow ZnCl_2 + H_2O$ (1); $ZnO + 2\,NaOH \longrightarrow Na_2ZnO_2 + H_2O$ (1).	1+1+1+1+1

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- Storing sodium or potassium in water; they are stored under kerosene because they react violently with water and air.
- Confusing **roasting** (sulphide ore, excess air) with **calcination** (carbonate ore, limited air).
- Writing the product of iron with steam as Fe_2O_3; it is Fe_3O_4 and hydrogen gas is released.
- Thinking that all metals are hard solids and all non-metals are non-conductors; Na is soft, Hg is a liquid and graphite conducts.
- Saying that an ionic solid conducts electricity; it conducts only when molten or in aqueous solution.

- Writing that copper can displace hydrogen from dilute acids or iron from $\mathrm{FeSO_4}$; a less reactive metal cannot displace a more reactive one.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 3: **Metals and Non-metals** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** nickel and chromium	Resists corrosion.
2	**(b)** conc. HCl and conc. HNO_3 in 3 : 1 ratio	It dissolves gold and platinum.
3	**(c)** in excess of air	Sulphide ores are heated in air.
4	**(a)** zinc	Zinc protects iron.
5	**(b)** gold	Gold can be drawn into a very thin wire.
6	**(c)** in limited or no air	Used for carbonate ores.
7	**(c)** aluminium oxide	It forms a thin protective oxide layer.
8	**(a)** silver	Silver is the best conductor, followed by copper.
9	**(a)** weld railway tracks	$Fe_2O_3 + 2\,Al \longrightarrow Al_2O_3 + 2\,Fe + heat$.
10	**(d)** iodine	Iodine is a lustrous non-metal.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Homogeneous mixture of two or more metals (or metal + non-metal) (1); Brass and bronze (1).	1+1
12	Mg and Zn (or Fe) (1); Cu and Ag (1).	1+1
13	High melting and boiling points (1); Soluble in water and conduct in solution/molten state (1).	1+1
14	Forms a protective layer of Al_2O_3 (1); The layer prevents further reaction (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Strong electrostatic force between ions needs lots of heat (1); In solid state ions are held in fixed positions (1); Ions move freely in molten or aqueous state (1).	1+1+1
16	Painting/oiling: keeps out air and moisture (1); Galvanisation: zinc layer (1); Alloying (stainless steel): modifies properties (1).	1+1+1
17	Reaction is violent and exothermic (1); $2\,Na + 2\,H_2O \longrightarrow 2\,NaOH + H_2$ (1); Hydrogen catches fire from the heat (1).	1+1+1
	OR: Roasting: heating a sulphide ore in excess air (1); $2\,ZnS + 3\,O_2 \longrightarrow 2\,ZnO + 2\,SO_2$ (1); Calcination: heating a carbonate ore in limited air: $ZnCO_3 \longrightarrow ZnO + CO_2$ (1).	1+1+1
18	Homogeneous mixture of metals (1); To improve strength, hardness or resistance to corrosion (1); Brass (Cu, Zn); bronze (Cu, Sn) (1).	1+1+1
	OR: Alkali (metal hydroxide) (1); $Na_2O + H_2O \longrightarrow 2\,NaOH$ (1); Not all metal oxides dissolve; only those of sodium, potassium, calcium (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Low (Hg, Cu): heating of sulphide ore (1); $2\,HgS + 3\,O_2 \longrightarrow 2\,HgO + 2\,SO_2$; $2\,HgO \longrightarrow 2\,Hg + O_2$ (1); Medium (Zn, Fe, Pb): roasting/calcination then carbon reduction (1); $ZnO + C \longrightarrow Zn + CO$ (1); High (Na, Al): electrolytic reduction of molten salt/oxide (1).	1+1+1+1+1
	OR: They are very reactive, above carbon in the series (1); Electrolysis of molten NaCl/Al_2O_3 (1); Cathode: $Na^+ + e^- \longrightarrow Na$ (1); Anode: $2\,Cl^- \longrightarrow Cl_2 + 2\,e^-$ (1); Carbon cannot reduce their oxides (1).	1+1+1+1+1
20	Anode: impure copper block (1); Cathode: strip of pure copper (1); Electrolyte: acidified $CuSO_4$ (1); Anode: $Cu \longrightarrow Cu^{2+} + 2\,e^-$; cathode: $Cu^{2+} + 2\,e^- \longrightarrow Cu$ (1); Impurities settle as anode mud (1).	1+1+1+1+1
	OR: $Fe_2O_3 + 2\,Al \longrightarrow Al_2O_3 + 2\,Fe$ (1); $2\,Na + Cl_2 \longrightarrow 2\,NaCl$ (1); $Mg + 2\,HCl \longrightarrow MgCl_2 + H_2$ (1); Hydrogen gas burns with a pop (1); Reactive metals displace hydrogen from acids (1).	1+1+1+1+1

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